EMBEDDED SYSTEMS & IoT DEVICE DESIGN

Task 2: Smart Light Control System Using LDR Sensor

Electronics & Communication Engineering

Maincraft Technologies Training Program

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Project Report

Smart Light Control System Using LDR Sensor

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Contents

1 Objective

The objective of this project is to design and simulate a Smart Light Control System using an LDR (Light Dependent Resistor) and Arduino UNO. The system automatically controls an LED based on ambient light intensity.

2 Introduction

Automation plays an important role in modern embedded systems. Smart lighting systems help reduce energy consumption by automatically switching lights ON or OFF according to surrounding light conditions.

This project demonstrates the interfacing of an LDR sensor with Arduino UNO to create an automatic light control system.

3 Components Used

S.No.	Comp onent
1	Arduino UNO
2	LDR (Light Dependent Resistor)
3	LED
4	10kΩ Resistor
5	220Ω Resistor
6	Breadb oard
7	Jumper Wires

4 Theory of LDR Sensor

An LDR (Light Dependent Resistor) is a sensor whose resistance changes according to the intensity of light falling on it.

4.1 Characteristics

- High resistance in darkness
- Low resistance in bright light
- Analog output suitable for Arduino ADC
- Low-cost and easy to use

4.2 Working

The LDR forms a voltage divider circuit with a resistor. The output voltage varies according to light intensity and is read by Arduino through Analog Pin A0.

5 Circuit Design

The LDR and 10kΩ resistor form a voltage divider circuit. The junction point is connected to Analog Pin A0 of Arduino UNO.

The LED is connected to Digital Pin 2 through a 220Ω resistor.

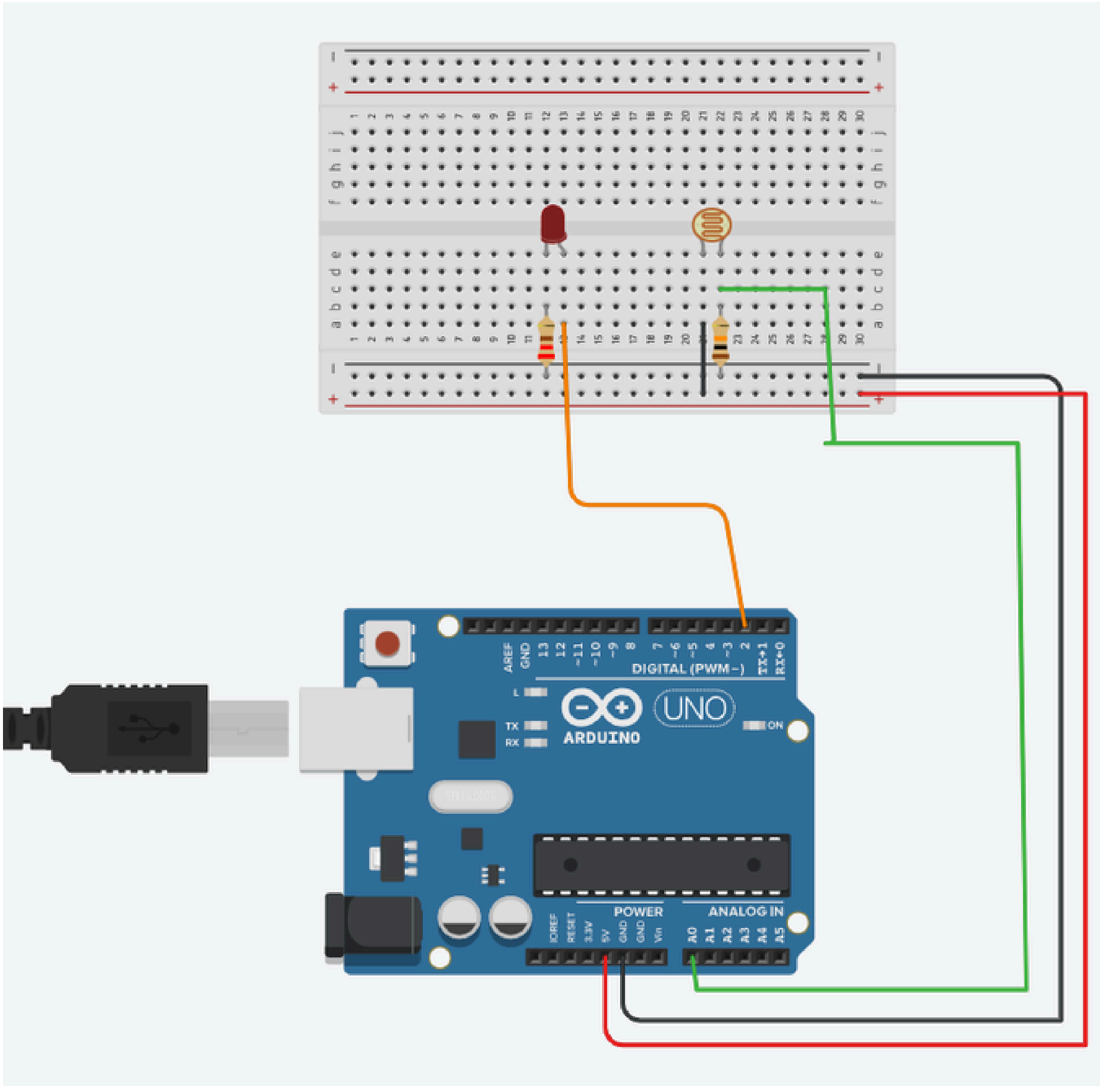

Figure 1: Smart Light Control System using Arduino UNO and LDR Sensor

6 WorkingPrinciple

The Arduino continuously reads the analog value from the LDR sensor.

- Low Light Condition → LED ON
- High Light Condition → LED OFF

Threshold Logic:

- LDR Value < 500 → LED ON
- LDR Value ≥ 500 → LED OFF

7 ArduinoProgram

```
// Smart Light Control System using LDR

int ldrPin = A0;
int ledPin = 2;

void setup() {
  pinMode(ledPin, OUTPUT);
  Serial.begin(9600);
}

void loop() {

  int ldrValue = analogRead(ldrPin);

  Serial.print("LDR Value = ");
  Serial.println(ldrValue);

  if (ldrValue < 500) {
    digitalWrite(ledPin, HIGH);
    Serial.println("LED ON (Dark Environment)");
  }
  else {
    digitalWrite(ledPin, LOW);
    Serial.println("LED OFF (Bright Environment)");
  }

  Serial.println("-------------------");

  delay(1000);
}
```

8 Simulation Results

8.1 Complete Circuit

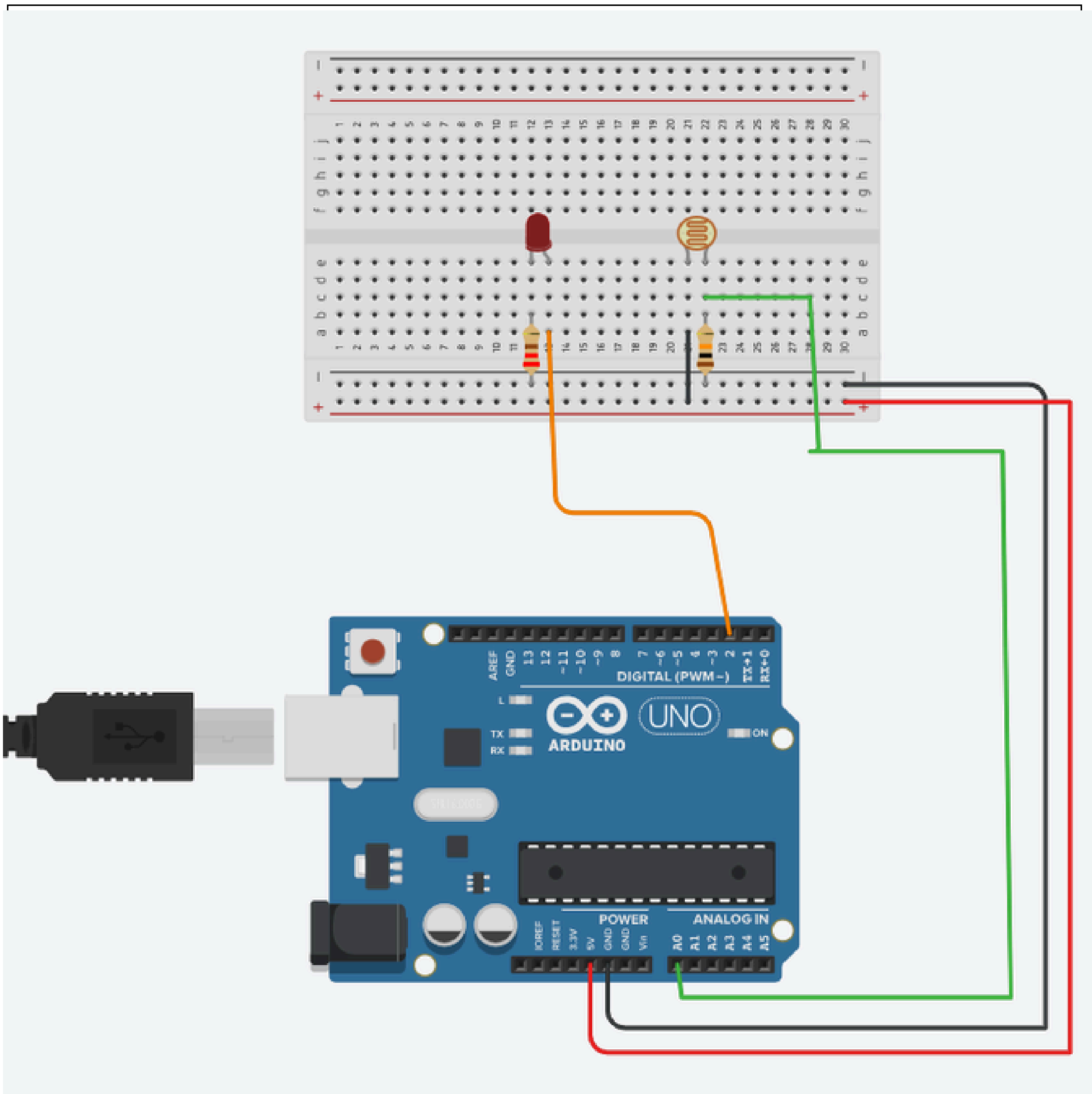

Figure 2: Complete Circuit Diagram

8.2 Low Light Condition

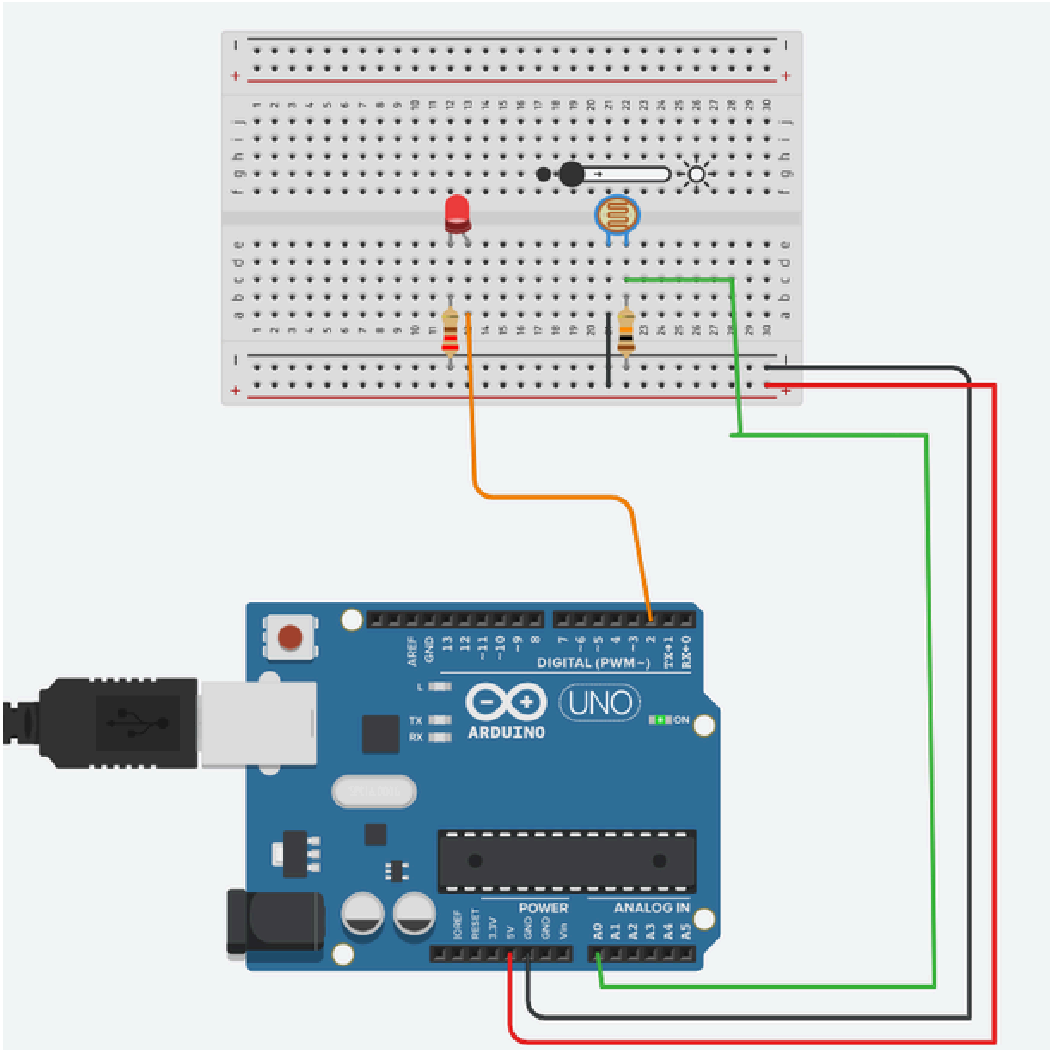

Figure 3: LDR Value 54, LED ON

8.3 HighLightCondition

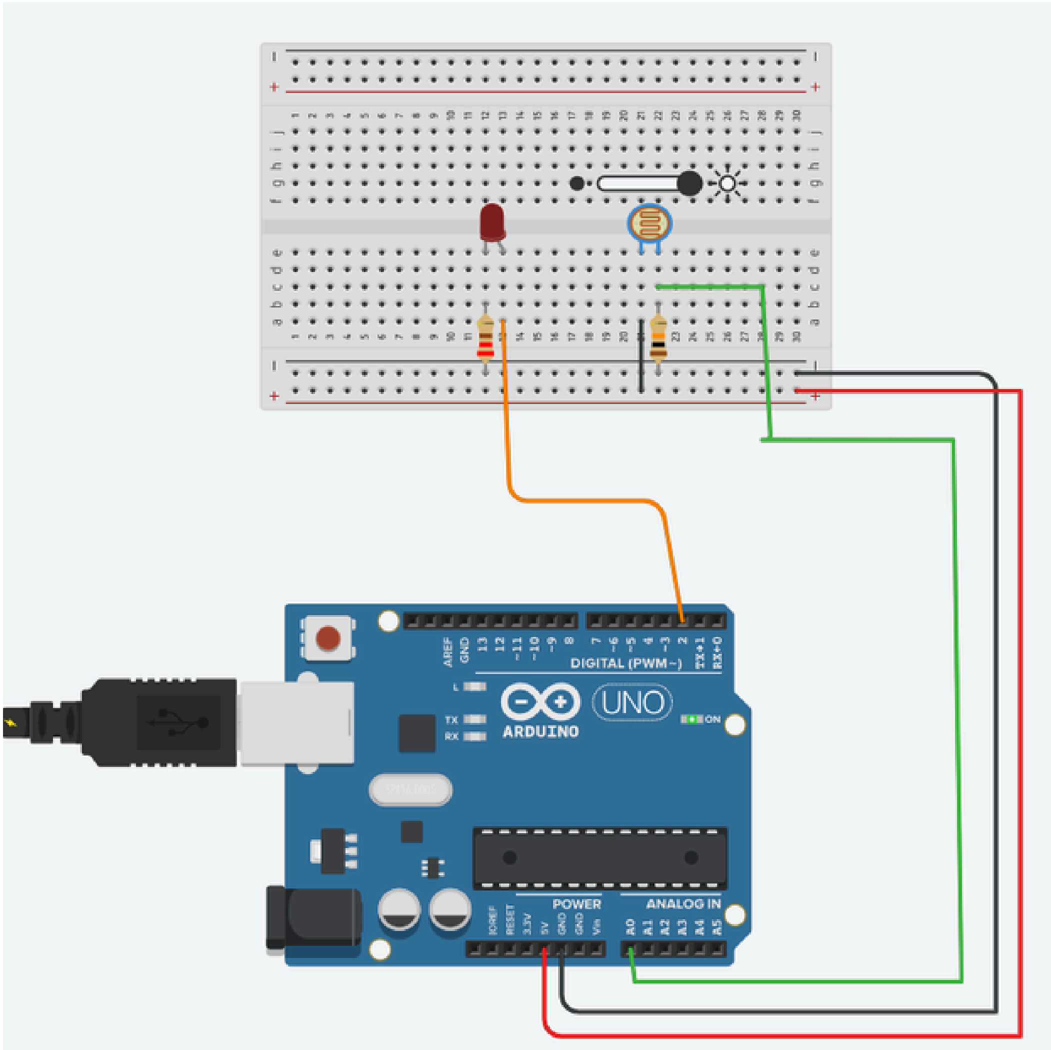

Figure 4: LDR Value 954, LED OFF

8.4 Serial Monitor Output

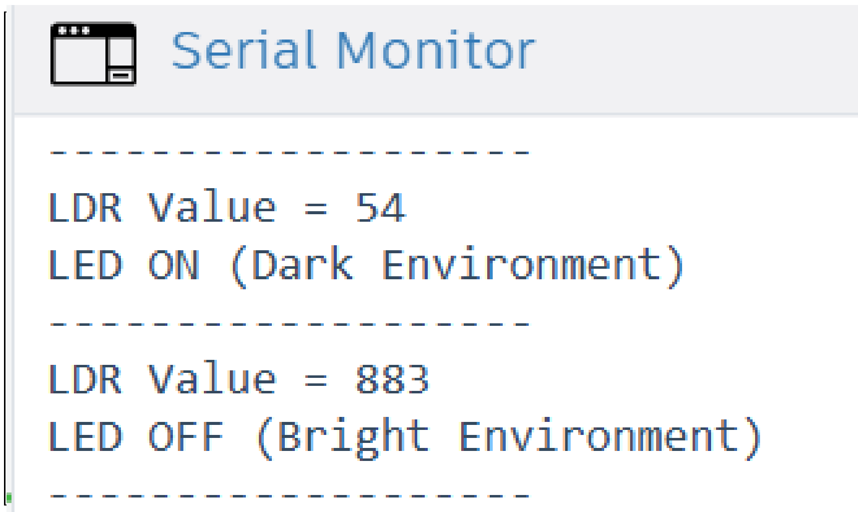

Figure 5: Real-Time LDR Values Displayed on Serial Monitor

9 Applications

- Automatic Street Lighting Systems
- Smart Home Automation
- Energy Saving Lighting Systems
- Industrial Lighting Control
- Garden Lighting Systems
- IoT-Based Smart Buildings

10 Advantages

- Low Cost Implementation

- Easy to Design and Maintain
- Low Power Consumption
- Automatic Operation
- Energy Efficient
- Suitable for Smart City Applications

11 Conclusion

The Smart Light Control System was successfully designed and simulated using Arduino UNO and an LDR sensor. The system automatically controls an LED based on ambient light intensity. The project demonstrates sensor interfacing, analog signal processing and embedded programming concepts.

12 References

1. Arduino Official Documentation
2. Tinkercad Circuits Simulator
3. LDR Sensor Datasheet
4. Arduino UNO Datasheet
5. Embedded Systems and IoT Training Material